Started on	Tuesday, 8 November 2022, 9:57 AM
State	Finished
Completed on	Tuesday, 8 November 2022, 10:00 AM
Time taken	2 mins 58 secs
Grade	**8.00** out of 10.00 (**80**%)

Question **1**

Correct

Mark 2.00 out of 2.00

How do we deallocate a variable in the .stack section?

Select one:

- a. We can't, as we need to wait to go out of scope. ✔
- b. We can't, as they are constant and kept in memory forever.
- c. We can, as the stack is writable.
- d. We can, as the .stack section can be resized with `brk`/`sbrk` calls.

The stack is a fixed memory region, the stack frames are moved around during execution.
This means that, though we can't deallocate the memory, we can ensure that we use variables only in the context they are needed.

The correct answer is: We can't, as we need to wait to go out of scope.

Question **2**

Correct

Mark 2.00 out of 2.00

How much memory is available for a process in a 64bit system?

Select one:

- a. ~2^64 bytes of memory. ✔
- b. ~4GB of memory.
- c. As much memory as we have.
- d. ~64GB of memory.

Physical memory is transparent to processes.
All processes think they have access to the whole memory in the address space.

The correct answer is: ~2^64 bytes of memory.

Question **3**

Incorrect

Mark 0.00 out of 2.00

What is the role of the TLB?

Select one:

- a. It plays a role similar to the CPU cache.
- b. It guarantees the existance of pages in memory. ✖
- c. It's part of the CPU architecture.
- d. It's part of the HDD.

The Translation Look-aside Buffer is similar to the cache in the CPU.
It maps the last accessed pages in order to make additional accesses faster.

The correct answer is: It plays a role similar to the CPU cache.

Question **4**

Correct

Mark 2.00 out of 2.00

How does the system identify which system call is performed?

Select one:

- a. A numeric value, written in `rax`. ✔
- b. A pointer to a string, written in `rdx`.
- c. The first four letters of the system call name, written in `rbx`.
- d. The name of the wrapper function used to perform the system call, written in `rax`.

All system calls have an numeric identifier that is specified in the `rax` register.
The system call's arguments are passed in `rdi`, `rsi`, `rdx`, `r10`, `r8`, `r9`,
in this order.

The correct answer is: A numeric value, written in `rax`.

Question **5**

Correct

Mark 2.00 out of 2.00

What information about a process can we obtain using `strace`?

Select one:

- a. What system calls it performs. ✔
- b. What library calls it performs.
- c. How many functions it calls.
- d. If it contains memory leaks.

`strace` is used to track what system calls the process performs.

The correct answer is: What system calls it performs.

◄ Curs 04: Date: Recap

Jump to...

Curs 06: Calcul (partea a 2-a) ►

Started on	Tuesday, 8 November 2022, 9:56 AM
State	Finished
Completed on	Tuesday, 8 November 2022, 9:57 AM
Time taken	1 min 44 secs
Grade	**10.00** out of 10.00 (**100**%)

Question **1**

Correct

Mark 2.00 out of 2.00

What information about a process can we obtain using `ltrace`?

Select one:

- a. What library calls it performs. ✔
- b. What system calls it performs.
- c. How many functions it calls.
- d. If it contains memory leaks.

`ltrace` is used to track what library calls the process performs.

The correct answer is: What library calls it performs.

Question **2**

Correct

Mark 2.00 out of 2.00

When is a reference to an object incremented?

Select one:

- a. When we assign the object to another object. ✔
- b. When we free the object.
- c. When we reinitialize the object.
- d. When we assign the object to itself.

Reference Counting does exactly what it's name says.
The system holds information on how many objects refer to another object/memory zone.

The correct answer is: When we assign the object to another object.

Question **3**

Correct

Mark 2.00 out of 2.00

How does the system identify which system call is performed?

Select one:

- a. A numeric value, written in `rax`. ✔
- b. A pointer to a string, written in `rdx`.
- c. The first four letters of the system call name, written in `rbx`.
- d. The name of the wrapper function used to perform the system call, written in `rax`.

All system calls have an numeric identifier that is specified in the `rax` register.
The system call's arguments are passed in `rdi`, `rsi`, `rdx`, `r10`, `r8`, `r9`, in this order.

The correct answer is: A numeric value, written in `rax`.

Question **4**

Correct

Mark 2.00 out of 2.00

What does a `mutable` variable refer to?

Select one:

- a. A modifiable variable ✔
- b. A moveable variable
- c. A readable variable
- d. An executable variable

`mutable` refers to altering (mutating) the content of a variable.

The correct answer is: A modifiable variable

Question **5**

Correct

Mark 2.00 out of 2.00

What happens with D class objects that go out of scope?

Select one:

- a. They are freed by the Garbage Collector. ✔
- b. They remain allocated like in C.
- c. D does not support classes.
- d. D class objects never freed in D.

D class objects are handled by the Garbage Collector.
This tracks the number of references to the object.
Once the garbage collector notices that there are no more references to that object, the memory used for it is freed.

The correct answer is: They are freed by the Garbage Collector.

◄ Curs 04: Date: Recap

Jump to...

Curs 06: Calcul (partea a 2-a) ►

Started on	Tuesday, 8 November 2022, 9:53 AM
State	Finished
Completed on	Tuesday, 8 November 2022, 9:55 AM
Time taken	2 mins 31 secs
Grade	**10.00** out of 10.00 (**100**%)

Question **1**

Correct

Mark 2.00 out of 2.00

How can we allocate memory in the `.heap` section in Python?

Select one:

- a. We do not handle memory zones in Python. ✔
- b. We use the `malloc` family equivalent in Python.
- c. We use the `-h` flag when starting Python.
- d. We use the `h` flag when allocating memory.

The type and size of memory zones is resolved at runtime in Python.
This means we have no control on the allocation using Python.

The correct answer is: We do not handle memory zones in Python.

Question **2**

Correct

Mark 2.00 out of 2.00

How do cache misses influence our program?

Select one:

- a. They add latency to read/write operations. ✔
- b. They corrupt our memory.
- c. They lead to reading the wrong data.
- d. They lead to writing the wrong data.

A cache miss implies that the data needs to be read/written from/to RAM.
This results in additional CPU cycles wasted on reading/writing data, which could be used to continue the execution.

The correct answer is: They add latency to read/write operations.

Question **3**

Correct

Mark 2.00 out of 2.00

How do we deallocate a variable in the .stack section?

Select one:

- a. We can't, as we need to wait to go out of scope. ✔
- b. We can't, as they are constant and kept in memory forever.
- c. We can, as the stack is writable.
- d. We can, as the .stack section can be resized with `brk`/`sbrk` calls.

The stack is a fixed memory region, the stack frames are moved around during execution.
This means that, though we can't deallocate the memory, we can ensure that we use variables only in the context they are needed.

The correct answer is: We can't, as we need to wait to go out of scope.

Question **4**

Correct

Mark 2.00 out of 2.00

What happens when we access a memory zone for which we have no rights?

Select one:

- a. A `Segmentation Fault` error is received. ✔
- b. A `Page Fault` warning is received.
- c. An `Access Restricted` error is received.
- d. The operation fails silently.

Because we don't have that specific type of access rights to the page, we get a `Segmentation Fault` error.
The access fails and the kernel will send a SIGSEGV signal to our process which will make it stop.

The correct answer is: A `Segmentation Fault` error is received.

Question **5**

Correct

Mark 2.00 out of 2.00

What happens with D class objects that go out of scope?

Select one:

- a. They are freed by the Garbage Collector. ✔
- b. They remain allocated like in C.
- c. D does not support classes.
- d. D class objects never freed in D.

D class objects are handled by the Garbage Collector.
This tracks the number of references to the object.
Once the garbage collector notices that there are no more references to that object, the memory used for it is freed.

The correct answer is: They are freed by the Garbage Collector.

◄ Curs 04: Date: Recap

Jump to...

Curs 06: Calcul (partea a 2-a) ►

Started on	Tuesday, 8 November 2022, 10:59 AM
State	Finished
Completed on	Tuesday, 8 November 2022, 11:03 AM
Time taken	4 mins 13 secs
Grade	**4.00** out of 10.00 (**40**%)

Question **1**

Incorrect

Mark 0.00 out of 2.00

What is the role of the TLB?

Select one:

- a. It plays a role similar to the CPU cache.
- b. It guarantees the existance of pages in memory. ✖
- c. It's part of the CPU architecture.
- d. It's part of the HDD.

The Translation Look-aside Buffer is similar to the cache in the CPU.
It maps the last accessed pages in order to make additional accesses faster.

The correct answer is: It plays a role similar to the CPU cache.

Question **2**

Incorrect

Mark 0.00 out of 2.00

Can we write to a `const` variable on the *stack*?

Select one:

- a. Yes, but we need to cast the variable first.
- b. Yes, we just asign a new value like `x = 3`.
- c. No, constant variables can never be modified.
- d. No, because the page permission is read-only. ✖

Even though the variable is marked with `const`, it's put in a writable section (the stack).
By using casting we can trick the compiler to only give a warning (no error).
Thus, we can write in the const zone.

The correct answer is: Yes, but we need to cast the variable first.

Question **3**

Correct

Mark 2.00 out of 2.00

What is the role of the MMU?

Select one:

- a. Checks whether virtual addresses are valid. ✔
- b. Handles reading/writing to disk.
- c. Guarantees coherent data is written to disk.
- d. It's a software component of the OS.

The Memory Management Unit is a hardware component of the memory.
It ensures that virtual addresses are valid when accessed by the CPU.
When a page fault happens, it is handled by the MMU.

The correct answer is: Checks whether virtual addresses are valid.

Question **4**

Correct

Mark 2.00 out of 2.00

How many `Segmentation Fault` errors are generated when accessing element 100 in a 10-element list in Python?

Select one:

- a. 0 ✔
- b. 1
- c. 90
- d. 99

Lists in Python are checked before being accessed.
If an incorrect index is accessed, it will generate an `Index Out of Bounds` error in Python.
The access is not made so no `Segmentation Fault` errors are generated.

The correct answer is: 0

Question **5**

Incorrect

Mark 0.00 out of 2.00

Which loads faster between a dynamic and a static executable?

Select one:

- a. A dynamic executable
- b. A static executable ✖
- c. They load in the same time frame.
- d. Loading does not depend on the executable type.

Dynamic executables have a smaller size, and, as such, are loaded faster.
Static executables include all the library code, which needs to be brought into memory first.

The correct answer is: A dynamic executable

Started on	Tuesday, 8 November 2022, 11:03 AM
State	Finished
Completed on	Tuesday, 8 November 2022, 11:07 AM
Time taken	4 mins 15 secs
Grade	**8.00** out of 10.00 (**80**%)

Question **1**

Correct

Mark 2.00 out of 2.00

When are system calls required?

Select one:

- a. When performing privileged operations. ✔
- b. When calling library functions.
- c. When copying strings.
- d. When the system notifies the process that an event has happened.

System calls are a mechanism to switch context from the userspace to the kernel space in order to perform privileged operations. Most operations that may impact other parts of the system (e.g. memory allocation, reading and writing I/O data, creating a thread or a process) require system calls. Regular computations, function calls and memory access within the process, including libraries, do not require system calls.

The correct answer is: When performing privileged operations.

Question **2**

Correct

Mark 2.00 out of 2.00

What happens when we read memory from an expired stack frame in C?

Select one:

- a. The access is valid, but the data inside might or might not be altered in the meantime. ✔
- b. The access is valid and we retrieve our data.
- c. The access results in a `Segmentation Fault` error.
- d. The access results in a `Stack Smashing` error.

Even though we have access to the memory zone of the old stack frame, the memory inside might have been overwritten with another stack frame.
We cannot trust the data from expired stack frames.

The correct answer is: The access is valid, but the data inside might or might not be altered in the meantime.

Question **3**

Incorrect

Mark 0.00 out of 2.00

Can a process access the memory of another process?

Select one:

- a. Yes, by using shared memory zones.
- b. Yes, by directly accessing the memory.
- c. Yes, by making changes in the physical memory.
- d. No, there is no such mechanism. ✖

The most common way to access memory is through shared memory zones.
Our program can use `mmap` to map virtual memory in a shared manner to share with other processes.
The communication between the two virtual memory zones is guaranteed by the OS.

The correct answer is: Yes, by using shared memory zones.

Question **4**

Correct

Mark 2.00 out of 2.00

What happens with D class objects that go out of scope?

Select one:

- a. They are freed by the Garbage Collector. ✔
- b. They remain allocated like in C.
- c. D does not support classes.
- d. D class objects never freed in D.

D class objects are handled by the Garbage Collector.
This tracks the number of references to the object.
Once the garbage collector notices that there are no more references to that object, the memory used for it is freed.

The correct answer is: They are freed by the Garbage Collector.

Question **5**

Correct

Mark 2.00 out of 2.00

What are the permissions of the .rodata process memory zone?

Select one:

- a. r _ _ ✔
- b. r w _
- c. r _ x
- d. r w x

The .rodata zone contains the initialized and constant variables from our program.
It can be seen as an extension of the .data section, but with different permissions.
In order to work with these variables we need to be able to read them.

The correct answer is: r _ _

◄ Curs 04: Date: Recap

Jump to...

Curs 06: Calcul (partea a 2-a) ►